Technological University of the Philippines Visayas
COLLEGE OF ENGINEERING
Talisay City, Negros Occidental

State Diagram Implementation with CPLD

A Project presented to the

Computer Engineering Department

of Technological University of the Philippines Visayas

In Partial Fulfillment

of the Requirements for the Subject

HDL

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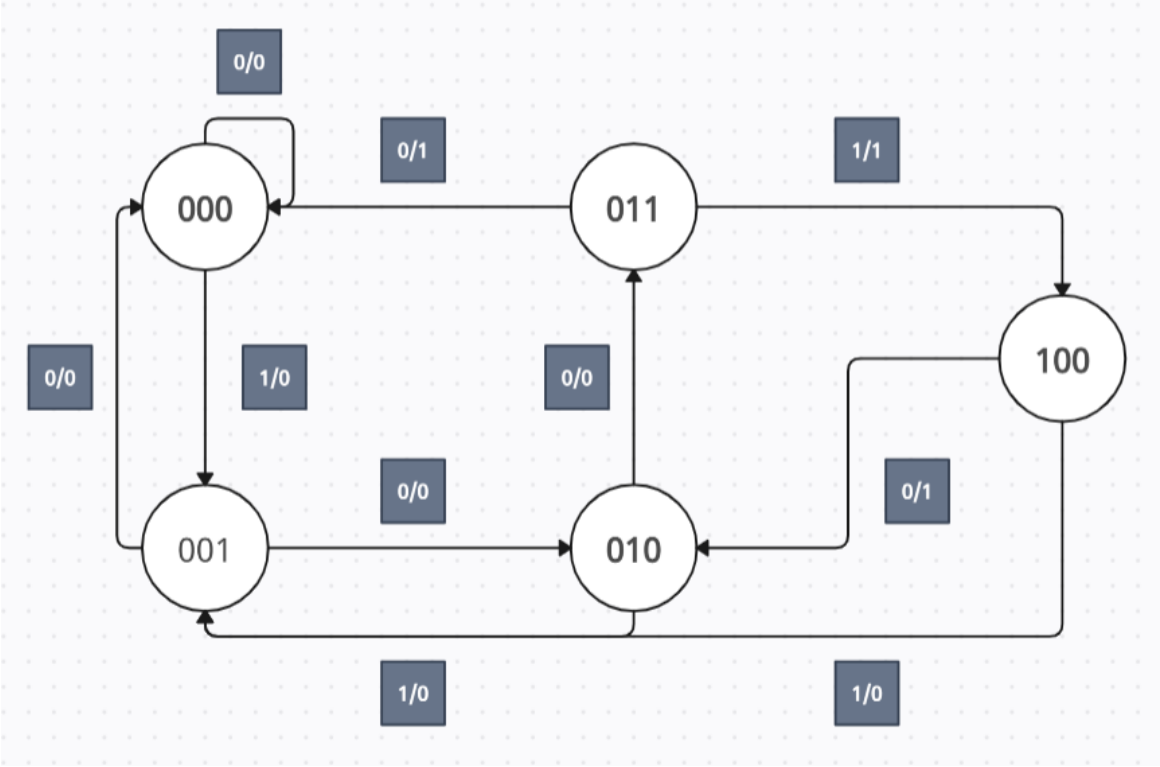
STATE DIAGRAM ANSWER



MODELSIM SIMULATION

Verilog Code Testbench

```
module activity_1_tb;
    reg clk, x;
    wire y, clk_led;

    activity_1 uut(clk, x, y, clk_led);

    initial begin
        clk = 0; x = 0;
        forever #2 clk = ~clk;
    end

    initial begin
        $display("Time\tInput\tOutput");

        #1 x = 1;  #2 $display("%0t\t%b\t%b", $time, x, y);
        #2 x = 0;  #2 $display("%0t\t%b\t%b", $time, x, y);
        #2 x = 0;  #2 $display("%0t\t%b\t%b", $time, x, y);
        #2 x = 1;  #2 $display("%0t\t%b\t%b", $time, x, y);
        #2 x = 0;  #2 $display("%0t\t%b\t%b", $time, x, y);
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        #2 x = 1;  #2 $display("%0t\t%b\t%b", $time, x, y);
        #2 x = 0;  #2 $display("%0t\t%b\t%b", $time, x, y);
        #2 x = 1;  #2 $display("%0t\t%b\t%b", $time, x, y);

        #1 $finish;
    end
endmodule
```
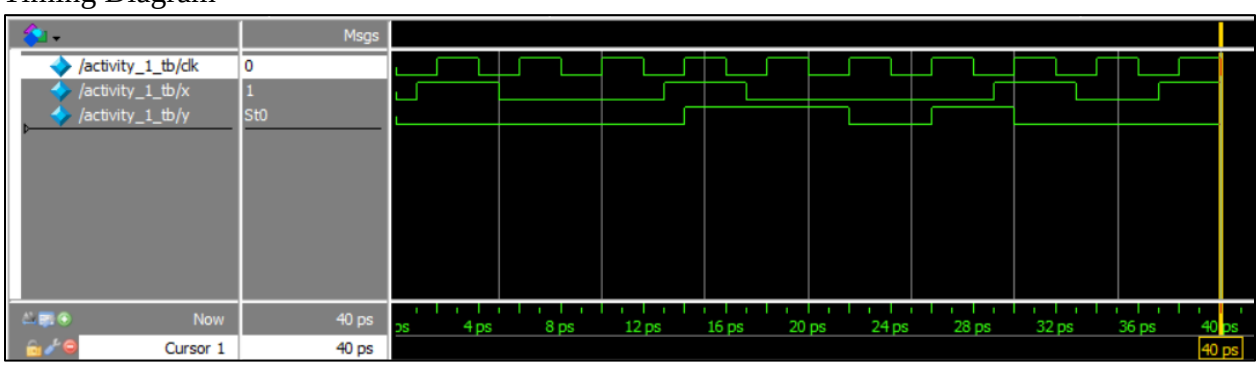

Timing Diagram



QUARTUS

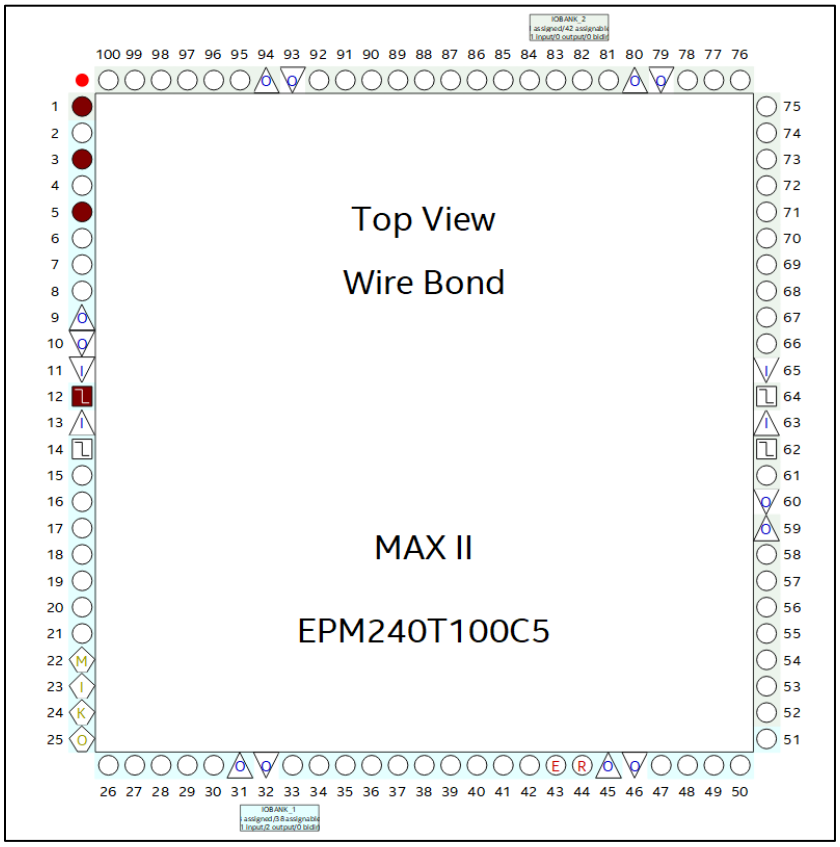
Verilog Code

```
module Activity_1(clk,x,y,clk_led);
   input clk, x;
   output y, clk_led;
   wire clk_slow;

      clock_divider slow_clk(clk,clk_slow);
      state_dia main(clk_slow,x,y);

      assign clk_led = clk_slow;

endmodule
```

```

module state_dia(clk, x, y);
   input clk, x;
   output reg y;

   reg[2:0] state, next_state;

   initial begin
      y = 0;
      state = 3'b00;
      next_state = 3'b00;
   end

   always @(posedge clk) begin

      case(state)
         3'b000:  begin
            if(x) next_state = 3'b001;
            else next_state = 3'b000;
            y = 0;
         end

         3'b001:  begin
            if(x) next_state = 3'b000;
            else next_state = 3'b010;
            y = 0;
         end

         3'b010:  begin
            if(x) next_state = 3'b001;
            else next_state = 3'b011;
            y = 0;
         end

         3'b011:  begin
            if(x) next_state = 3'b100;
            else next_state = 3'b000;
            y = 1;
         end

         3'b100:  begin
            if(x) next_state = 3'b001;
            else next_state = 3'b010;
            y = ~x;
         end

      endcase

      state <= next_state;
   end

endmodule
```

```
module clock_divider(clk_in,clk_out);

    input clk_in;
    output reg clk_out;

    reg [26:0] counter;

    initial begin
        counter = 0;
        clk_out = 0;
    end

   always @(posedge clk_in) begin
      counter <= counter + 1;
       if (counter > 50000000) begin
           counter <= 0;
           clk_out <= ~clk_out;
       end
   end
endmodule
```

Pin Planner



Node Name	Direction	Location	I/O Bank	Fitter Location	I/O Standard	Reserved	Current Strength	Slew Rate Preservation
clk	Input	PIN_12	1	PIN_14	3.3-V ...fault)		16mA ...ault)	
clk_led	Output	PIN_5	1	PIN_6	3.3-V ...fault)		16mA ...ault)	
x	Input	PIN_1	2	PIN_68	3.3-V ...fault)		16mA ...ault)	
y	Output	PIN_3	1	PIN_69	3.3-V ...fault)		16mA ...ault)	
<<new node>>								

Flow Summary

Flow Summary

<<Filter>>

Flow Status	Successful - Wed Jan 22 00:02:32 2025
Quartus Prime Version	20.1.0 Build 711 06/05/2020 SJ Lite Edition
Revision Name	Activity_1
Top-level Entity Name	Activity_1
Family	MAX II
Device	EPM240T100C5
Timing Models	Final
Total logic elements	44 / 240 (18 %)
Total pins	4 / 80 (5 %)
Total virtual pins	0
UFM blocks	0 / 1 (0 %)